

FretGarden

Pedagogical Constitution

Chapter 10

Curriculum Architecture, Sequencing, and Prerequisite Graphs

Version	1.0
Status	Foundational Draft
Document Type	Normative Pedagogical Standard
Authority	Binding on curriculum maps, sequences, prerequisite graphs, pathways, units, integration points, placement routes, and curriculum-level progression decisions
Primary Dependencies	Foundation; Chapters 2 through 9
Scope	The architecture by which FretGarden organizes learning over time without reducing musicianship to a single linear course

This chapter defines how FretGarden transforms isolated objectives into a coherent, evidence-based, revisitable, and musically integrated curriculum.

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Chapter 10 - Curriculum Architecture, Sequencing, and Prerequisite Graphs

Version: 1.0

Status: Foundational Draft

Document Type: Normative Pedagogical Standard

Authority: Subordinate to the Foundation and binding on all curriculum maps, sequence specifications, prerequisite graphs, pathways, unit structures, placement routes, and curriculum-level progression decisions.

Primary Dependencies: Foundation Sections 2, 4, 6, 7, 8, 9, 10, and 13; Chapters 2 through 9.

1. Purpose

This chapter defines how FretGarden organizes learning across lessons, units, pathways, and time. It establishes the architecture by which individual learning objectives become a coherent curriculum rather than a collection of accurate but disconnected activities.

The chapter governs curriculum sequencing, prerequisite relationships, branching, spiral return, integration points, placement, bypass, review obligations, remediation routes, and the coordination of parallel domains such as fretboard knowledge, rhythm, ear training, conceptual reasoning, physical execution, musical application, and practice independence.

A curriculum must do more than decide what comes first. It must explain why an objective is available, what prior capabilities make it learnable, what evidence permits advancement, how forgotten material re-enters the path, where separate domains must converge, and how learner choice can exist without hiding dependencies or weakening standards.

This chapter does not prescribe the complete FretGarden topic sequence, final course catalog, universal lesson count, visual curriculum map, or implementation data model. It establishes the constitutional rules those later specifications must follow.

2. Constitutional Authority

This chapter operationalizes the Foundation requirement that FretGarden develop durable, transferable musicianship through patient, focused, musically meaningful practice. It also coordinates the objective, evidence, lesson, exercise, assessment, mastery, review, feedback, and remediation systems defined in Chapters 2 through 9.

Where a curriculum map conflicts with an objective definition, evidence requirement, readiness rule, review obligation, remediation requirement, or constitutional commitment, the higher pedagogical requirement governs. A sequence may not be defended solely because it is conventional, marketable, easy to implement, familiar to instructors, or efficient to display.

Temporary curriculum noncompliance **MUST** be documented as educational debt. The record **MUST** identify the affected objectives, learner risk, interim mitigation, and condition for correction.

3. Governing Principle

FretGarden *MUST* organize learning as an evidence-aware network of capabilities whose sequence reflects genuine dependencies, whose pathways permit purposeful variation, and whose repeated integrations move the learner toward independent musical use.

The curriculum is coherent when each learner action can be justified by the capability being developed, the evidence currently available, the dependencies of the next demand, and the long-term musical purpose of the path.

4. Core Distinctions

4.1 Curriculum Is Not a Topic List

A topic list names content. A curriculum specifies capabilities, dependencies, instructional routes, evidence obligations, review returns, and integration. “Intervals, scales, chords, modes” is a topic list. A curriculum explains what learners must do with each, how one capability supports another, and how the knowledge becomes audible, playable, retrievable, and musically usable.

4.2 Sequence Is Not Mere Order

Sequence is an evidence-based relationship among demands. Two objectives may appear in an order without the first being necessary for the second. Conversely, a prerequisite may need to be reactivated immediately before use even when it was taught months earlier. Sequence therefore includes dependency, timing, spacing, and readiness conditions.

4.3 Prerequisite Is Not Prior Exposure

A learner has not satisfied a prerequisite merely by seeing a lesson, completing a unit, or once producing a correct answer. A prerequisite is satisfied only when current evidence supports the level of capability required by the next objective.

4.4 Completion Is Not Readiness

Completion records participation or passage through content. Readiness is a claim about present capability relative to a future demand. Curriculum systems **MUST** preserve this distinction even when the interface allows forward movement.

4.5 Linear Is Not Coherent

A strictly linear path may be simple to display yet educationally incoherent if it delays necessary integration, ignores uneven development, or prevents appropriate return. Coherence comes from explicit dependencies and integration, not from a single unbroken list.

4.6 Choice Is Not Absence of Structure

Learner agency requires meaningful options within known constraints. It does not require removing prerequisites, presenting all content as equally appropriate, or concealing the educational consequences of a choice.

4.7 Integration Is Not Juxtaposition

Placing rhythm, theory, fretboard, and ear tasks in the same unit does not establish integration. Integration requires the learner to coordinate those capabilities in one musically meaningful demand.

4.8 Difficulty Is Not Level

A curriculum level represents the scope and integration of capability, not simply task hardness. A fast but narrow drill may be more difficult than a slow integrated task while still representing less advanced musicianship.

5. Definitions

Curriculum: A designed system of objectives, dependencies, instructional routes, evidence requirements, review obligations, and musical applications organized over time.

Curriculum architecture: The structural rules by which objectives are grouped, connected, revisited, integrated, and made available.

Objective graph: A directed, typed representation in which nodes are scoped learning objectives or integration objectives and edges describe educational relationships.

Node: A defined capability, integration target, review obligation, placement decision, or other curriculum-relevant state with explicit evidence conditions.

Edge: A documented relationship between nodes, including prerequisite, co-requisite, supporting, enrichment, transfer, review, remediation, or integration relationships.

Foundational spine: The shared set of capabilities required across most long-term FretGarden pathways.

Strand: A connected line of development within a domain or cross-domain capability family.

Pathway: A purposeful route through the graph oriented toward a learner goal while preserving required dependencies.

Integration point: A node or sequence requiring coordinated use of capabilities previously developed in separate strands.

Spiral return: A planned revisit that increases scope, independence, variation, integration, or musical consequence rather than merely repeating prior instruction.

Placement: A process for locating the learner within the graph using defensible evidence of current capability.

Bypass: Permission to omit instruction or practice because evidence already supports the relevant objective and its required conditions.

Exploratory access: Permission to encounter later material without treating access as evidence of readiness or mastery.

Return-to-path: The explicit route from remediation, reactivation, or optional exploration back to ordinary progression.

Curriculum debt: A documented mismatch between the constitutional curriculum requirement and the currently implemented curriculum.

6. Curriculum Contract

Every approved curriculum pathway MUST identify:

- its intended learner population and musical purpose;
- the objectives and objective domains it develops;
- the entry conditions and placement assumptions;
- the prerequisite and co-requisite relationships that govern availability;
- the evidence required for advancement, bypass, and revalidation;
- the required review and spaced-retrieval obligations;
- the integration points that connect separate domains;
- the permitted learner choices and the consequences of those choices;
- the remediation and return-to-path routes;
- the exit capabilities and claim limits;
- and the version of the curriculum graph under which learner records were produced.

A pathway that cannot state these elements is a content collection, not a constitutionally complete curriculum.

7. Curriculum Architecture Overview

FretGarden curriculum architecture consists of five interacting layers:

1. The objective layer defines the capabilities to be developed.
2. The dependency layer defines what those capabilities require and support.
3. The pathway layer defines purposeful routes through the graph.
4. The evidence layer governs availability, advancement, review, and revalidation.
5. The integration layer ensures that isolated capabilities become coordinated musicianship.

No layer may be designed in isolation. A sequence that ignores evidence becomes completion-driven. An evidence system without pathways becomes administratively precise but educationally directionless. A pathway without integration produces fragmented skill. Integration without prerequisite control creates overload and invalid failure.

8. Objective Graph as the Primary Structure

The authoritative curriculum representation **SHOULD** be an objective graph rather than a flat ordered list. Learner-facing views may display courses, gardens, maps, paths, units, or recommendations, but those views **MUST** remain projections of the underlying graph.

The graph **MUST** permit multiple valid routes while preserving required dependencies. It **MUST** also represent review, reactivation, remediation, and integration edges rather than treating them as exceptions outside the curriculum.

A flat sequence **MAY** be used for a constrained beginner experience when it accurately reflects the relevant graph segment. It **MUST NOT** become the hidden basis for all later curriculum decisions.

9. Nodes

A curriculum node **MUST** be small enough to support interpretable evidence and meaningful dependency relationships, but large enough to represent a capability rather than a single screen or arbitrary task.

Each core node **MUST** include at minimum: objective identifier; objective statement; domain classification; scope; conditions; quality standard; independence requirement; expected time horizon; prerequisite set; evidence requirement; review obligation; and permitted next states.

Nodes **MAY** represent cross-domain integration, but an integration node **MUST** name the component capabilities that are causally necessary for success. It may not use “integration” as a vague label for a large mixed activity.

10. Edges and Dependency Types

Edges **MUST** be typed. An undifferentiated “comes before” relationship is insufficient because different educational relationships produce different availability and progression rules.

Permitted edge types include:

- required prerequisite: the target is not instructionally valid without the source capability;
- strongly recommended prerequisite: the target may be attempted, but insufficient source capability is likely to create avoidable difficulty;
- supporting prerequisite: the source improves efficiency or understanding but is not causally required;
- co-requisite: capabilities are expected to develop together and must be coordinated before later advancement;
- enrichment: the source deepens or broadens learning without controlling core progression;
- transfer: the target requires application of the source under meaningfully changed conditions;
- integration: the target requires simultaneous coordination of multiple source capabilities;
- review: the target rechecks or strengthens the source after delay;
- reactivation: the target restores accessibility of a previously evidenced capability;
- remediation: the target repairs a barrier preventing the ordinary path;
- and equivalence: alternate nodes support an equivalent claim under approved conditions.

Every edge **MUST** state its rationale and the evidence condition under which it is considered satisfied.

11. Prerequisite Classification

Prerequisites **MUST** be classified according to the demand imposed by the target objective, not according to tradition or instructor preference. A capability is required only when failure to possess it would make the target invalid, unsafe, uninterpretable, or predictably overloaded.

Prerequisite classification **MUST** identify the relevant dimension. A learner may need accurate note retrieval but not fluent note retrieval; stable pulse but not advanced subdivision; conceptual recognition but not full verbal explanation. Requiring more than the target actually needs creates unnecessary delay. Requiring less creates hidden overload.

A prerequisite relationship **MUST** be reviewed when the target lesson repeatedly succeeds without it or repeatedly fails despite it.

12. Prerequisite Sufficiency

Prerequisite sufficiency is target-relative. Evidence sufficient for one target may be insufficient for another. For example, recognition of interval names may support an introductory comparison lesson but not real-time chord-tone navigation.

A prerequisite check **MUST** consider accuracy, independence, recency, scope, and transfer conditions relevant to the target. Old evidence **MAY** support historical achievement while still requiring reactivation before a high-demand integration.

The system **MUST** distinguish absent evidence, stale evidence, contradictory evidence, and demonstrated weakness. These states require different actions.

13. Readiness at the Graph Level

Graph-level readiness is the degree to which the learner currently satisfies the dependency profile of a target node or pathway segment.

A readiness decision **MUST** produce one of the states defined by Chapter 6, such as ready, ready with support, exploratory access, review recommended, remediation required, readiness uncertain, or temporarily inaccessible.

Readiness **MUST NOT** be reduced to one aggregate score. A learner may be conceptually ready but temporally unready, or fretboard-ready but perceptually underprepared. The next action should address the limiting dimension rather than obscure it within an average.

14. Core, Supporting, and Enrichment Objectives

Curriculum objectives **MUST** be classified by their role in the larger system.

- Core objectives are required for the stated pathway or for the shared foundational spine.
- Supporting objectives improve comprehension, efficiency, expression, or transfer but do not independently block progression unless the target explicitly depends on them.
- Enrichment objectives extend style, repertoire, depth, or curiosity without being misrepresented as universal requirements.

The classification **MAY** vary by pathway. Reading standard notation may be supporting in one pathway and core in another. The curriculum record **MUST** therefore identify the classification within a defined pathway rather than treating it as globally permanent.

15. Foundational Spine

FretGarden **MUST** maintain a shared foundational spine sufficient to support meaningful progress across major guitar and music-theory pathways. The spine **MUST** not become an exhaustive gate that delays all choice until every foundational topic is mastered.

The spine **SHOULD** include appropriately scoped capabilities in orientation, sound-pitch connection, note and interval understanding, fretboard navigation, pulse and subdivision, basic chord and scale relationships, deliberate practice, retrieval, and self-correction.

Spine objectives **MUST** be selected because they unlock broad future learning, not because they are conventionally taught first. Each must have explicit downstream dependencies or mission-level value.

16. Parallel Development Strands

FretGarden **SHOULD** develop major capability families in parallel rather than completing one domain before beginning another. A learner should not finish all conceptual theory before hearing, playing, navigating, or applying it.

Parallel strands may include conceptual reasoning, ear training, fretboard knowledge, rhythm, physical execution, musical application, and metacognitive practice. Their pacing may differ, but the curriculum **MUST** create planned synchronization points.

Parallel development reduces brittle dependence on one representation and allows early musical meaning. It also creates a need for careful load control: the number of simultaneously novel demands **MUST** remain within the lesson and curriculum novelty budget.

17. Branching Pathways

Branching pathways allow learners to pursue different musical goals while sharing a coherent foundation. Valid branches may emphasize rhythm guitar, improvisation, songwriting, fretboard fluency, ear development, theory depth, bass adaptation, or genre-specific application.

Every branch **MUST** declare: shared prerequisites; branch-specific objectives; optional enrichment; integration points with the core; and conditions for returning to or crossing into another branch.

Branches **MUST** not become isolated silos. A branch may change emphasis and examples, but it may not silently weaken mission-level commitments to sound, timing, understanding, transfer, and sustainable practice.

18. Spiral Progression

FretGarden **MUST** use spiral progression when a capability benefits from repeated encounters at increasing levels of scope, independence, variation, or integration.

A spiral return is valid only when it changes at least one meaningful demand. Repeating the same explanation or exercise set without changed conditions is review or repetition, not spiral progression.

Spiral design **SHOULD** make prior learning causally useful. Later work should retrieve and transform earlier capability rather than merely mention it.

19. Integration Points

Integration points are mandatory where the curriculum claims musicianship rather than isolated task performance. They require the learner to coordinate capabilities under musically meaningful constraints.

An integration point **MUST** define component objectives, observable dimensions, support conditions, and claim limits. Failure must be diagnosable: the system should determine whether the barrier is conceptual, perceptual, navigational, temporal, physical, attentional, or integrative.

Examples include locating and playing chord tones in time over a progression, hearing and reproducing a scale-degree pattern from multiple roots, or explaining and applying an interval relationship within a short musical phrase.

20. Units, Modules, and Courses

Units, modules, and courses are organizational containers. They are not themselves evidence of learning. Their boundaries **SHOULD** correspond to coherent capability clusters, integration goals, or meaningful learner decisions.

A unit **MUST** identify its core objectives, entry conditions, internal sequence logic, exit evidence, delayed return obligations, and relationship to other units. A course **MUST** additionally identify its pathway purpose and final capability profile.

Container completion **MAY** be recorded for orientation and motivation, but it **MUST** remain separate from the current status of constituent objectives.

21. Curriculum Granularity

Curriculum granularity **MUST** support authoring, evidence, review, and progression without overwhelming learners or maintainers.

Objectives that are too broad conceal uneven capability and make remediation vague. Objectives that are too narrow create administrative noise and fragment musical meaning. The correct granularity is the smallest unit for which FretGarden can define meaningful evidence, dependency, review, and application.

Related micro-capabilities **MAY** be grouped into a curriculum node when they share dependencies, evidence conditions, and review behavior. They **MUST** remain separate when one can be absent while the other is present in a way that changes instruction or progression.

22. Sequence Construction Rules

A curriculum sequence **MUST** be constructed by working backward from intended capability and forward from learner entry conditions.

6. Define the pathway exit capabilities and musical uses.
7. Decompose them into objective nodes at appropriate granularity.
8. Identify required and supporting dependencies.
9. Locate integration points and transfer demands.
10. Specify evidence and review obligations.
11. Design plausible learner routes, including placement and remediation.
12. Test the sequence for cognitive load, motivational sustainability, and hidden prerequisites.
13. Audit whether each earlier objective is used later and whether each later demand is prepared adequately.

Sequence authors **MUST** document major alternatives considered when conventional order conflicts with evidence, learner experience, or implementation constraints.

23. Entry Conditions

Every pathway and substantial unit **MUST** state its entry conditions. Entry conditions **MAY** include prior capability, instrument access, tuning, physical safety assumptions, notation familiarity, audio availability, or practice-time expectations.

An entry condition **MUST** be necessary or materially relevant. It may not be included merely because prior users usually possessed it.

When an entry condition cannot be assumed, the curriculum **MUST** provide placement, orientation, an equivalent route, or transparent limitation.

24. Exit Conditions

Exit conditions **MUST** define what evidence permits FretGarden to claim that the learner is prepared for specified next demands. They **MUST** distinguish lesson completion, unit completion, functional readiness, retained capability, and broader mastery.

An exit condition **SHOULD** be multidimensional and may require delayed evidence, varied roots or contexts, unsupported retrieval, temporal stability, or musical application. The condition must remain proportional to the consequence of the claim.

No unit exit may claim mastery solely from immediate post-instruction performance.

25. Placement and Bypass

Placement **MUST** recognize prior learning without weakening evidence standards. The learner should not be forced through instruction for capabilities already demonstrated, nor advanced on the basis of self-report alone when the downstream consequence is significant.

Bypass evidence MUST sample the capability under conditions representative of future use. Efficient screening may begin with brief tasks, but stronger claims require stronger evidence when uncertainty remains.

A bypass decision MUST preserve review obligations. Skipping instruction does not make the capability permanent or exempt from later revalidation.

26. Exploratory Access

FretGarden MAY permit exploratory access to later material when doing so supports curiosity, goal formation, or musical meaning without creating false progression claims.

Exploratory access MUST be labeled honestly. The learner MUST be told which prerequisites may limit the experience, which results are not being interpreted as mastery, and how to return to the recommended path.

Exploration MUST NOT silently alter mastery records or unlock dependent certification claims.

27. Advancement With Support

Advancement with support is permitted when the target can be learned validly while one or more prerequisites remain below fully independent status, provided the support is explicit, recorded, and planned for fading.

Support may include references, reduced scope, slower tempo, worked examples, guided choices, or co-requisite practice. It may not conceal that the prerequisite remains unstable.

The system MUST define the point at which support dependence becomes a remediation trigger or blocks further advancement.

28. Review Within Curriculum Architecture

Review is part of curriculum architecture, not a separate queue attached after content is authored. Every core objective MUST have a planned return path after initial instruction.

Review nodes MAY occur within later lessons, dedicated sessions, integration tasks, or natural musical use when the target capability is genuinely retrieved and observable.

Curriculum authors MUST prevent review debt from accumulating beyond a sustainable load. Review selection should prioritize prerequisite value, evidence recency, learner weakness, and upcoming demand.

29. Remediation and Return-to-Path

Every required prerequisite MUST have at least one remediation route. Remediation MAY branch from the ordinary path, but it MUST define the original barrier, changed conditions, evidence of repair, and return-to-path condition.

The return route SHOULD be as short as educationally defensible. Learners must not be trapped in a parallel remedial curriculum after the relevant capability has been restored.

Repeated failure after remediation MUST trigger reconsideration of task validity, accessibility, prerequisite classification, or the inferred barrier rather than endless repetition.

30. Lapse, Reactivation, and Revalidation

A lapsed capability remains part of the learner history but may no longer support current readiness. The graph MUST preserve historical achievement while routing the learner through reactivation or revalidation when needed.

Reactivation SHOULD be lighter than first-time instruction when evidence suggests partial retention. Re-teaching is required when the learner cannot reconstruct the capability with proportionate support.

Revalidation MUST occur under conditions relevant to the upcoming demand, not merely under the easiest familiar format.

31. Interleaving Across the Curriculum

Interleaving **MUST** be introduced when learners possess enough initial stability for contrast and selection to be productive. It should require the learner to identify which concept, strategy, location, or response applies rather than repeat one blocked procedure.

Interleaving **MAY** occur within lessons, across a practice session, or between curriculum strands. Its difficulty must be calibrated so that errors remain interpretable.

Random variety is not sufficient. Interleaved items **SHOULD** be selected because their distinctions matter for future musical use.

32. Transfer Architecture

Every major capability family **MUST** include planned near-transfer and far-transfer opportunities appropriate to its scope. Transfer cannot be assumed merely because the learner has completed varied exercises.

Near transfer changes incidental features such as root, string, register, representation, tempo, or starting point while preserving the central relationship. Farther transfer changes musical context, task format, or coordinated demand while retaining the underlying capability.

Transfer tasks **MUST** be sequenced after sufficient initial stability but before the learner becomes dependent on one familiar context.

33. Musical Application Architecture

Curriculum pathways **MUST** connect isolated capabilities to genuine musical decisions and performances. Musical application **SHOULD** appear early in constrained form and increase in openness as capability grows.

Application tasks **MAY** include riff construction, accompaniment, transcription, improvisation, analysis, ear-to-fretboard reproduction, chord-tone targeting, rhythmic transformation, or repertoire use.

Application **MUST** be evaluated against explicit constraints and objectives. Personal stylistic preference may not be treated as the only correctness standard.

34. Rhythm as a Cross-Curricular Requirement

Timing and pulse are not confined to a rhythm unit. Where a capability is intended for real-time musical use, the curriculum **MUST** eventually require performance in relation to pulse, subdivision, phrase length, or coordinated timing.

Tempo demands **MUST** be introduced only after the underlying action is sufficiently stable. Speed must not substitute for accuracy, relaxation, or understanding.

Curriculum graphs **SHOULD** represent temporal dependencies explicitly so that a learner is not judged incapable of a conceptual or navigational objective solely because an unrelated tempo demand was introduced prematurely.

35. Ear Training as a Cross-Curricular Requirement

Theory objectives that concern sound relationships **SHOULD** include perceptual grounding and later auditory retrieval where appropriate. Ear training must not remain an isolated elective if the pathway claims understanding of intervals, scale degrees, chord qualities, tonal function, or melodic relationships.

Perceptual prerequisites and conceptual prerequisites **MUST** remain distinguishable. A learner may name a relationship without hearing it or hear it without naming it; curriculum routes should develop and later integrate both.

36. Fretboard Navigation as a Cross-Curricular Requirement

Fretboard knowledge **MUST** progress beyond memorized diagrams. Curriculum pathways **SHOULD** require locating, constructing, and applying relationships across strings, roots, positions, and representations.

Shape familiarity **MAY** support early action but **MUST** not be treated as sufficient evidence of note, interval, scale-degree, or chord-tone understanding unless the learner must use the claimed relationship to succeed.

Navigation load must be calibrated separately from conceptual and temporal load so that failures remain diagnosable.

37. Physical Execution and Safety Dependencies

Physical technique prerequisites **MUST** be limited to what the target musically requires. Curriculum authors must avoid embedding unnecessary speed, stretch, endurance, or coordination demands inside conceptual assessments.

Safety-relevant dependencies, such as pain-free posture, manageable tension, or instrument setup, **MUST** override momentum. The system may recommend human instruction when automated guidance cannot evaluate the condition responsibly.

Equivalent demonstrations **SHOULD** be defined when the musical capability can be shown without the standard physical response mode.

38. Metacognitive and Practice Dependencies

Long-term progression depends on the learner's ability to plan, focus, observe, adjust, stop, rest, and return. These capabilities **MUST** be represented as curriculum objectives rather than treated only as motivational advice.

Practice independence **SHOULD** develop in parallel with musical content. Learners should increasingly identify the target, select a suitable tempo or task size, recognize error patterns, and choose a next action.

Pomodoro-style intervals **MAY** structure focused work, but curriculum architecture **MUST** remain useful for different sustainable session lengths.

39. Curriculum Load and Pacing

Pacing **MUST** account for novelty, integration demand, physical effort, review burden, and learner goals. It may not be determined solely by lesson count or average completion time.

The curriculum **SHOULD** offer short, meaningful progress units while preserving long-term coherence. A short session may advance one objective, reactivate a prerequisite, or complete a focused integration attempt without requiring an entire module.

The system **MUST** avoid both overload and artificial delay. Learners who demonstrate readiness should be permitted to move efficiently; learners who need more time should receive a better path rather than stigma.

40. Short-Session and Long-Term Coherence

Every short activity **SHOULD** connect visibly or structurally to a larger capability trajectory. The learner should be able to understand why the activity matters and what future use it supports.

Long-term paths **MUST** remain resumable after interruption. The system should identify stale prerequisites, prioritize reactivation, and restore context without forcing complete restart.

Curriculum progress representations **SHOULD** emphasize growing capability and maintained knowledge rather than uninterrupted streaks or raw volume alone.

41. Learner Choice and Agency

Learners SHOULD be offered meaningful choices in examples, genres, application tasks, optional branches, practice order within safe bounds, and review selection where core obligations remain satisfied.

Choice MUST be informed. The system should explain why a recommendation exists, what dependency is unresolved, and what consequence follows from taking an exploratory or alternate route.

Learner preference MAY influence route and representation but MUST not independently redefine the evidence required for a capability claim.

42. Multiple Goals and Learner Profiles

FretGarden MAY support multiple learner goals through different pathway projections over the shared objective graph. A beginner seeking fretboard fluency, a songwriter seeking harmonic understanding, and a bassist seeking interval navigation may share objectives while differing in emphasis, examples, and integration demands.

Pathway personalization MUST preserve transparent claim scope. Mastery within one tuning, instrument, genre constraint, or task family may not be generalized automatically beyond the evidence.

Profiles SHOULD influence recommendations rather than permanently lock learners into a route.

43. Curriculum States and Status Language

Learner-facing curriculum states MUST distinguish available, recommended, ready with support, exploratory, review due, remediation required, completed, evidenced, retained, lapsed, and optional where those distinctions affect decisions.

Status language MUST remain encouraging without inflating capability. “Completed” should describe participation; “ready” should describe target-relative readiness; “mastered” should be reserved for evidence meeting the defined mastery claim.

The interface MAY simplify language, but the underlying record MUST preserve the constitutional distinctions.

44. Curriculum Evidence Records

Curriculum records MUST separate observations, inferred capability, confidence, support conditions, completion history, readiness decisions, review due states, and pathway recommendations.

Each progression decision SHOULD be traceable to the objective version, evidence considered, dependency rule, and decision policy in effect at the time.

Raw activity volume, time spent, streak length, or lesson completion MAY inform recommendations but MUST NOT independently establish readiness or mastery.

45. Curriculum Change Management

Changes to objective definitions, prerequisite edges, pathway structure, thresholds, or integration requirements MAY affect existing learner records. Such changes MUST be versioned and reviewed for backward compatibility.

A curriculum amendment MUST identify: the reason for change; affected nodes and edges; learner populations impacted; treatment of prior evidence; migration or revalidation rules; and any temporary educational debt.

A change that improves local convenience but damages cross-curriculum coherence MUST NOT be approved without explicit constitutional review.

46. Curriculum Anti-Patterns

14. A single locked sequence for all learner goals.

15. Unlocking later content solely because earlier screens were completed.

16. Treating every earlier topic as a required prerequisite.
17. Using one aggregate level to hide uneven capability.
18. Finishing all theory before allowing musical application.
19. Separating ear training, rhythm, and fretboard work so completely that they never integrate.
20. Repeating identical lessons and calling the result spiral learning.
21. Using harder or faster tasks as the only sign of advancement.
22. Allowing optional exploration to alter mastery records.
23. Creating remedial loops with no explicit return-to-path.
24. Ignoring stale evidence when a high-demand prerequisite is reused.
25. Treating branch-specific knowledge as universally required.
26. Overloading integration tasks with several unprepared novel demands.
27. Using genre examples as decoration rather than functional application.
28. Changing prerequisite graphs without migration or revalidation rules.
29. Optimizing curriculum order for implementation simplicity alone.

47. Constitutional Non-Negotiables

30. Every core curriculum objective **MUST** be defined according to Chapter 2.
31. Every required prerequisite **MUST** be justified by the target demand.
32. Completion **MUST** remain distinct from readiness and mastery.
33. The curriculum **MUST** support review, reactivation, remediation, and return-to-path.
34. Core pathways **MUST** include sound, timing, fretboard, conceptual, application, and practice integration appropriate to their claims.
35. Learner choice **MUST** not conceal dependencies or weaken evidence standards.
36. Branches **MUST** preserve mission-level educational commitments.
37. Integration points **MUST** require genuine coordination of component capabilities.
38. Placement and bypass **MUST** rely on defensible evidence.
39. Stale or contradictory evidence **MUST** be represented honestly.
40. Curriculum changes **MUST** be versioned and evaluated across the graph.
41. Every curriculum container **MUST** identify its objective, entry, exit, review, and dependency logic.
42. Short-session design **MUST** remain connected to long-term capability growth.
43. No sequence may be approved solely because it is conventional or easy to implement.
44. Curriculum architecture **MUST** develop increasing learner independence.

48. Conformance Tests

A curriculum design conforms only if reviewers can answer:

45. What capabilities does the path intend to produce?
46. What evidence supports each major progression decision?
47. Which prerequisites are truly required, and in which dimensions?
48. How are stale, missing, and contradictory prerequisite evidence handled?
49. Where do conceptual, perceptual, navigational, temporal, physical, and application strands develop?
50. Where are they integrated?
51. How does the path support different learner goals without losing coherence?
52. What choices can learners make, and what consequences are disclosed?
53. How can prior knowledge be placed or bypassed?
54. How does every core objective return after delay?
55. What remediation route exists for each blocking prerequisite?
56. How does the learner return to ordinary progression?
57. What prevents completion from being mistaken for capability?
58. How are short sessions connected to long-term development?
59. How are graph changes versioned and prior records treated?
60. Could any task failure be caused by an invalid hidden dependency or overload?
61. Does the path lead toward independent musical use rather than dependency on prompts?

49. Tradeoffs and Resolution Rules

49.1 Coherence Versus Choice

A shared structure supports cumulative learning; excessive constraint suppresses learner goals. Resolve the tension with a foundational spine, transparent branches, and explicit integration points rather than either total lockstep or unrestricted access.

49.2 Efficiency Versus Evidence

Fast unlock rules reduce friction but risk false readiness. Use efficient screening for low-consequence decisions and stronger evidence for high-consequence progression, bypass, or mastery claims.

49.3 Breadth Versus Depth

Broad exposure supports motivation and context; depth supports durable capability. Permit bounded exploration while protecting core retrieval, application, and review obligations.

49.4 Standardization Versus Personalization

Shared definitions and graph semantics preserve integrity. Personalize route, pacing, examples, and support while keeping objective meaning and evidence interpretation stable.

49.5 Momentum Versus Prerequisite Repair

Visible movement can sustain motivation, but unresolved required prerequisites compound difficulty. Preserve momentum through short targeted remediation, parallel safe work, and clear return-to-path rather than false advancement.

49.6 Simplicity Versus Accuracy

Learner-facing maps should be understandable, but the underlying model must preserve multidimensional states and typed dependencies. Simplify presentation, not educational meaning.

49.7 Stability Versus Curriculum Evolution

A stable path supports trust and longitudinal records; revision is necessary when evidence or experience reveals flaws. Version changes, migrate carefully, and revalidate only where the changed demand materially affects prior claims.

50. Assumptions Made

62. FretGarden will maintain identifiable learning objectives and evidence records.
63. The product will support more than one long-term learner goal.
64. Learners will progress unevenly across domains.
65. Some prior learning will be recognized through placement or bypass.
66. Some objectives will require instrument access and others can be practiced away from the instrument.
67. Review and reactivation will occur across months or years.
68. Curriculum views may differ from the authoritative underlying graph.
69. Automated recommendations will operate under uncertainty and must preserve claim limits.
70. Domain-specific chapters will define valid sequences and thresholds within this architecture.
71. The curriculum will evolve through versioned amendments rather than untracked replacement.

51. Unresolved Questions

72. The exact membership of the foundational spine.
73. Default beginner pathway length and unit boundaries.
74. Standard quantitative limits for active prerequisites.
75. Default policies for exploratory access to advanced material.
76. Graph visualization language for learners.

77. Confidence thresholds for automated placement and bypass.
78. Rules for cross-instrument and alternate-tuning evidence portability.
79. Minimum integration requirements by pathway.
80. How natural repertoire use should update objective evidence.
81. How branch changes should affect active review obligations.
82. Default migration rules after prerequisite-edge changes.
83. Governance of instructor override and learner-requested exceptions.
84. Accessibility-equivalent pathway approval.
85. The balance between universal core objectives and pathway-specific alternatives.
86. Methods for validating the curriculum graph empirically after release.

52. Topics Intentionally Deferred

87. The complete FretGarden curriculum map.
88. Final objective identifiers and graph schema.
89. Domain-specific lesson sequences.
90. Default mastery thresholds and timing windows.
91. Placement assessment item banks.
92. Learner-facing map and progress visualization.
93. Recommendation algorithms.
94. Educational analytics and graph validation metrics.
95. Accessibility implementation standards.
96. Repertoire and genre pathway catalogs.
97. Instructor authoring and override workflows.
98. Database and API representations.
99. Automated migration tooling for curriculum versions.
100. Formal exception, waiver, and educational-debt governance.

53. Required Future Cross-References

53.1 Practice-Session Architecture

Must define how curriculum recommendations become sustainable sessions, how review and new learning are balanced, and how learner-selected work remains connected to objective obligations.

53.2 Fretboard and Note-Learning Pedagogy

Must define objective nodes, prerequisite edges, representative sampling, spiral returns, and integration standards for note and fretboard capabilities.

53.3 Interval, Scale-Degree, and Chord-Tone Pedagogy

Must define conceptual, perceptual, navigational, and application strands and their integration points.

53.4 Rhythm and Metronome Pedagogy

Must define temporal prerequisites, tempo progression, pulse and subdivision strands, and cross-curricular timing requirements.

53.5 Ear-Training Pedagogy

Must define perceptual dependency structures, auditory transfer, guessing controls, and integration with theory and fretboard action.

53.6 Scales, Chords, Harmony, and Musical Application

Must define pathway branches, integration objectives, transfer contexts, and functional mastery requirements.

53.7 Accessibility

Must define equivalent demonstrations, alternate routes, and rules separating access barriers from musical capability.

53.8 Educational Analytics

Must distinguish graph state, raw evidence, inferred readiness, recommendation, intervention, and outcome.

53.9 Constitutional Governance

Must define review, amendment, versioning, migration, exception, waiver, and educational-debt procedures for curriculum architecture.

54. Chapter Conformance Statement

A FretGarden curriculum conforms to this chapter only when it organizes defined objectives through justified dependencies, evidence-aware progression, planned review, transparent choice, targeted remediation, and repeated musical integration.

A sequence is not constitutionally valid merely because it is orderly, comprehensive, familiar, or engaging. It must create a defensible path from learner entry conditions to durable and transferable capability while preserving honest status, sustainable practice, and increasing independence.

Curriculum architecture succeeds when learners do not merely move through content, but build a connected system of musical capabilities that remains available, adaptable, and usable beyond the path that first taught it.